

02LQD Specification and Simulation of Digital Systems

2022/06/27

1. Combinational circuit design (2 points)

Let X be the 1-bit Minuend, let Y be the 1-bit Subtrahend, let B_{in} be the 1-bit Borrow-in. X , Y and B_{in} are inputs. Let D be the 1-bit Difference, let B_{out} be the 1-bit Borrow-out. D and B_{out} are outputs. Design in VHDL a **full-subtractor** according to the following description styles:

- behavioural using IF THEN ELSIF statements
- behavioural using the CASE WHEN statement.

2. FSM design (16 points)

An FSM has a 1-bit input X and a 2-bit output Z . Its goal is to count how many 01 sequences appear in a 4-bit window. This value is available on Z once the 4th bit of the input sequence has been read, otherwise Z is at 0.

Example:

$X = 0101101011110001$

$Z_I = 0001000000000000$

$Z_0 = 0000000100000001$

1. Draw the state diagram of the Moore Machine (4 points)
2. Draw the state diagram of the Mealy Machine (4 points)
3. Design the **Moore** FSM resorting to the **VHDL** behavioural description style with 2 processes (4 points).
4. Design the **Mealy** FSM resorting in **Verilog** with 2 always blocks (4 points).

3. HLSM (12 points)

A RAM has 1024 16-bit cells. A functional unit has a 10-bit input $Addr$ and a 1-bit input Go as well as the signals (to be defined by the candidate) required to access the RAM. When the Go signal is asserted for 1 clock cycle, the functional unit reads a memory address on its $Addr$ input, then it counts how many bits are at 1 in the corresponding 16-bit cell in the RAM (*popcount*). Reset is synchronous. Design the functional unit as a HLSM, draw its state diagram (6 points) and write the VHDL code (6 points) to capture its behavior.

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

no later than Thursday June 30 14 5 pm

- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.